INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
SIM1006 - TUTORIAL 1

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1. State the order and degree of the following differential equations.

   (a) $x(y')^3 + y' = 8xy$ (c) $x^3 + yy' - (y')^5 = 0$

   (b) $\sqrt{y'} = 6(y'')^{2/3}$ (d) $(y'')^4 + xy(y')^6 + 7y = e^{2x}$

2. Show that $y = e^x - x$ is the explicit solution to $y' + y^2 = e^{2x} + (1 - 2x)e^x + x^2 - 1$.

3. Show that $y^2 + x - 3 = 0$ is an implicit solution to $\frac{dy}{dx} = -\frac{1}{2y}$ on the interval $(-\infty, 3)$.

4. Show that the given function $y(x)$ is the solution of the corresponding differential equation for any choice of the constants c_1 and c_2.

   (a) $y'' + y = 0$, $y(x) = c_1 \sin x + c_2 \cos x$.

   (b) $y'' - y' - 2y = -2x - 1$, $y(x) = c_1 e^{2x} + c_2 e^{-x} + x$

5. Show that the given function is the solution of the corresponding differential equation. Use implicit differentiation.

   (a) $x^2(y')^2 + xyy' - 2y^2 = 0$, $x^2 y = c$, c is constant.

   (b) $(\sin y)y' + \sin x \cos y = \sin x$, $e^{\cos x}(1 - \cos y) = c$, c is a constant.

6. Find the value of m such that the given function $y = e^{mx}$ is the solution to the differential equation $y''' + 3y'' + 2y' = 0$.

7. Show that the given function is the solution of the corresponding differential equation. Then, find the particular solution to the initial value problem.

   (a) $y' = y \cot x$, $y = c \sin x$, when $y = 2$, $x = \frac{\pi}{2}$.

   (b) $xy' = 3y$, $y = cx^3$, when $y = 4$, $x = -2$.

8. Verify that the function $y = c_1 e^x + c_2 e^{-2x}$ is a solution to $y'' + y' - 2y = 0$ for any choice of the constants c_1 and c_2. Determine c_1 and c_2 so that the initial conditions $y(1) = 1$ and $y'(1) = 0$ are satisfied.